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Distributed Consensus Protocols in Wireless Sensor Topologies

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Published Proceedings of the IEEE (2007)

Paper Link https://doi.org/10.1109/JPROC.2006.887293

Domain DOM_NET

Abstract

We review theoretical consensus algorithms for multi-agent network agreements. Using algebraic graph theory, we prove that asymptotic agreement across all network nodes is governed by the second smallest eigenvalue (Fiedler value) of the graph Laplacian.

Abstract quoted from R. Olfati-Saber et al., Proceedings of the IEEE (2007).

Notes: record the page or section for every claim you make about this paper.

Reading prompts

1. What is the core problem addressed and what assumptions are made?
2. What mathematical or structural abstractions represent the physical phenomena?
3. How can this concept be mapped or transferred to analogous problems in other domains?